Southeast University

Department of Computer Science and Engineering

Course Name: Database Design Lab

Course Code: CSE384

Lab No: 04

Lab Title: Basic SQL Queries with Attributes and Operators

Date of Submission: 01/12/2025

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Course: CSE384

Section: 07

Objective: Learn to retrieve and filter data using basic SELECT statements and operators like =, >, <, LIKE, and ORDER BY. And learn commands.

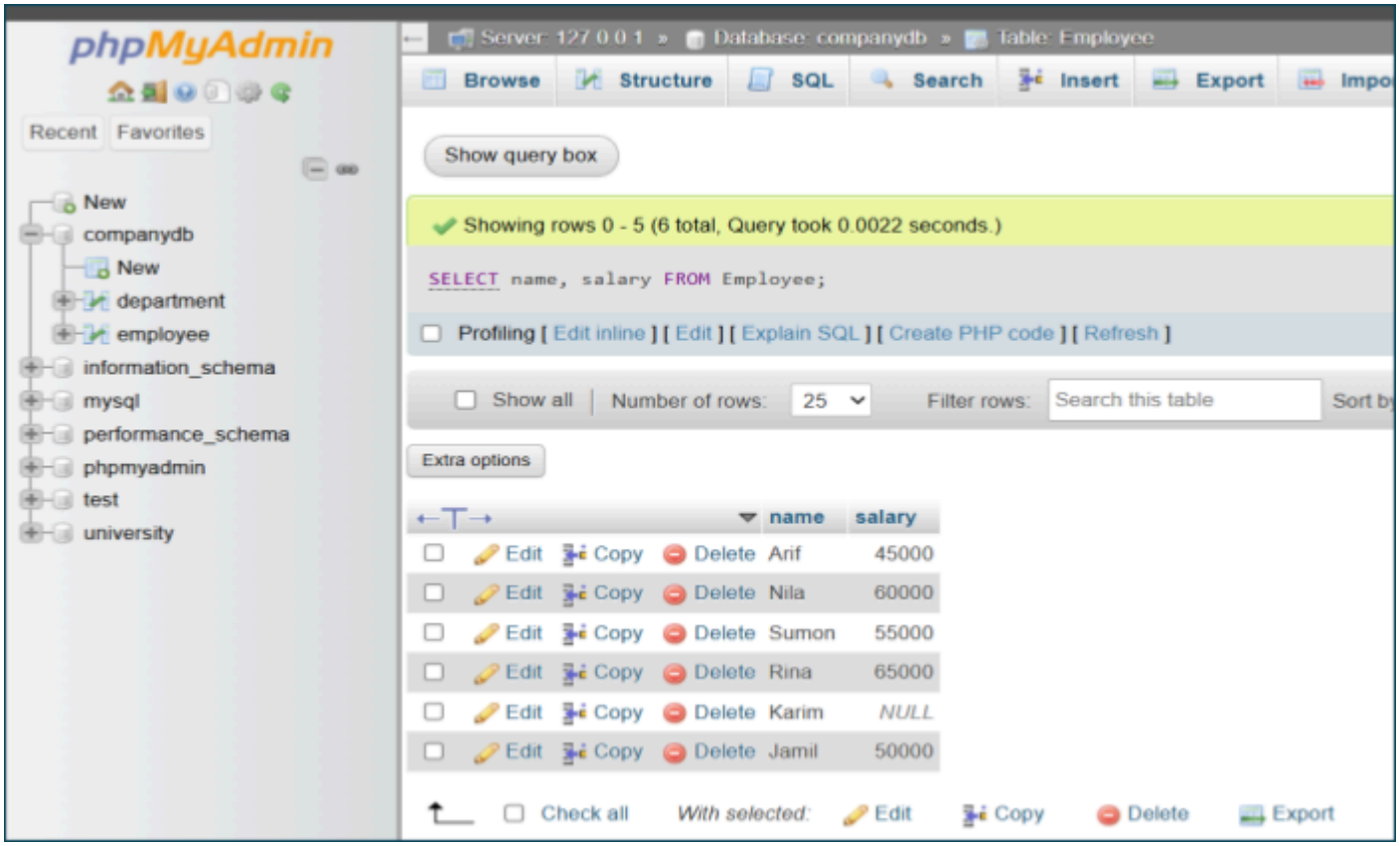
Problems and Solutions:

Problem 1:

Title: Show all employee names and their salaries.

SQL Query: `SELECT name, salary FROM Employee;`

Output:

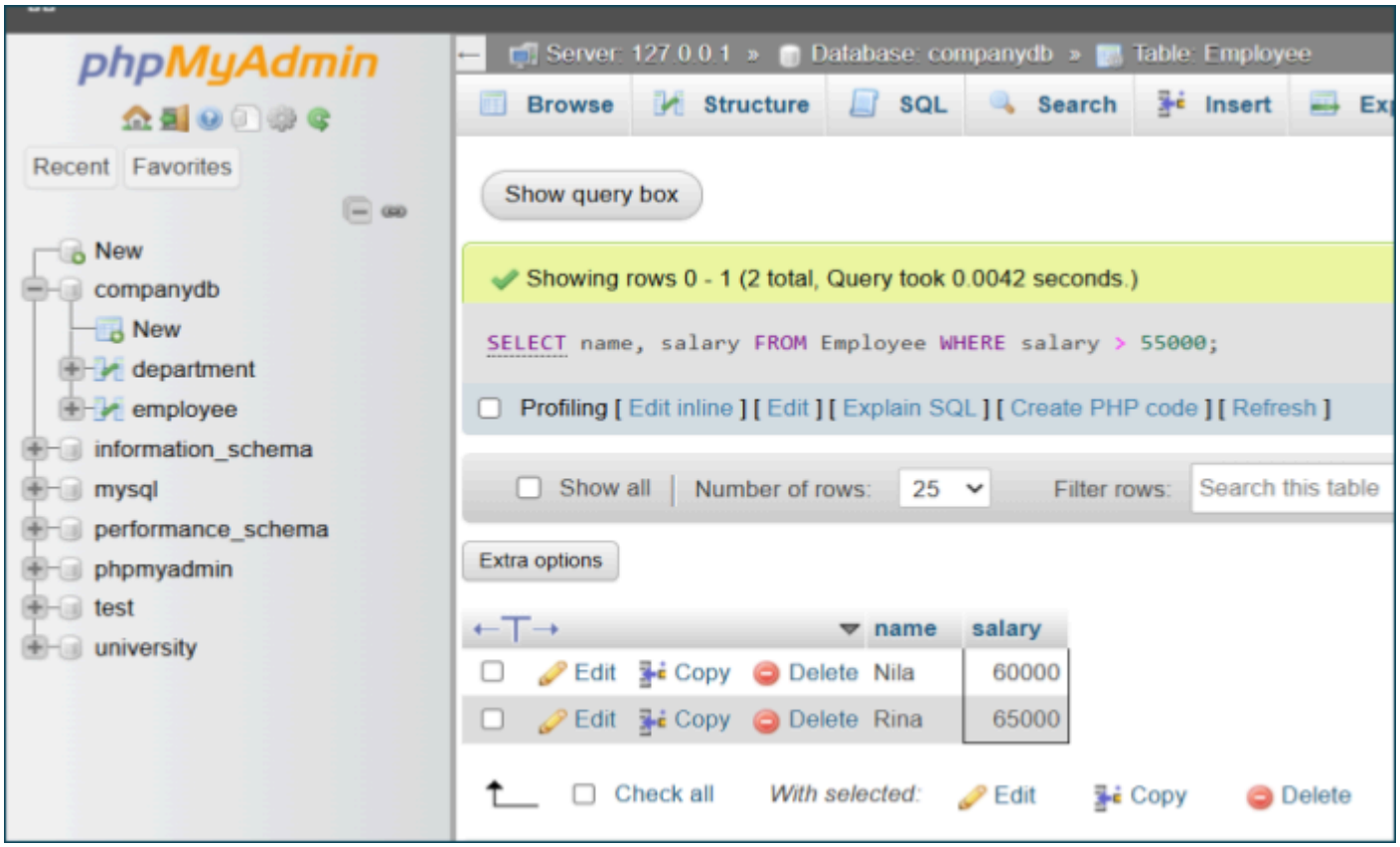


Problem 2:

Title: Find employees with a salary greater than 55000.

SQL Query: `SELECT name, salary FROM Employee WHERE salary > 55000;`

Output:

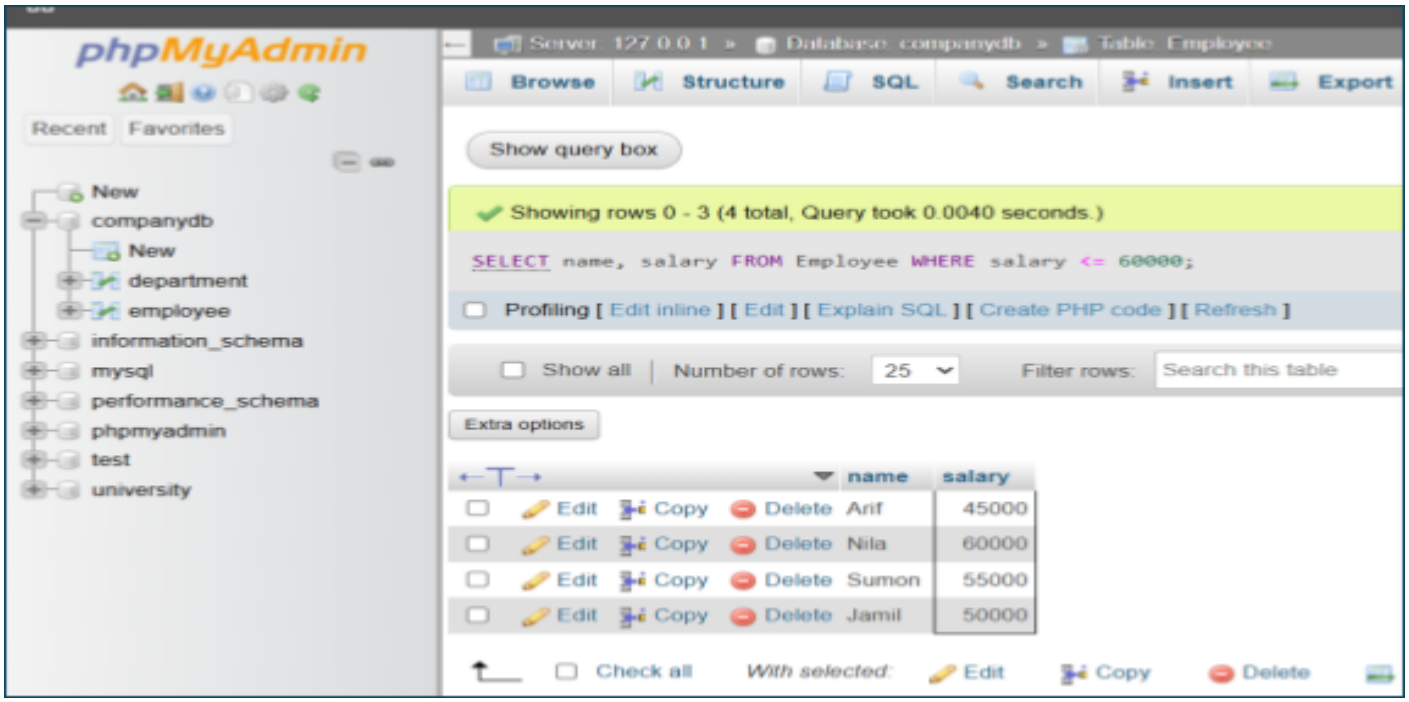


Problem 3:

Title: Find employees with salary less than or equal to 60000.

SQL Query: `SELECT name, salary FROM Employee WHERE salary <= 60000;`

Output:

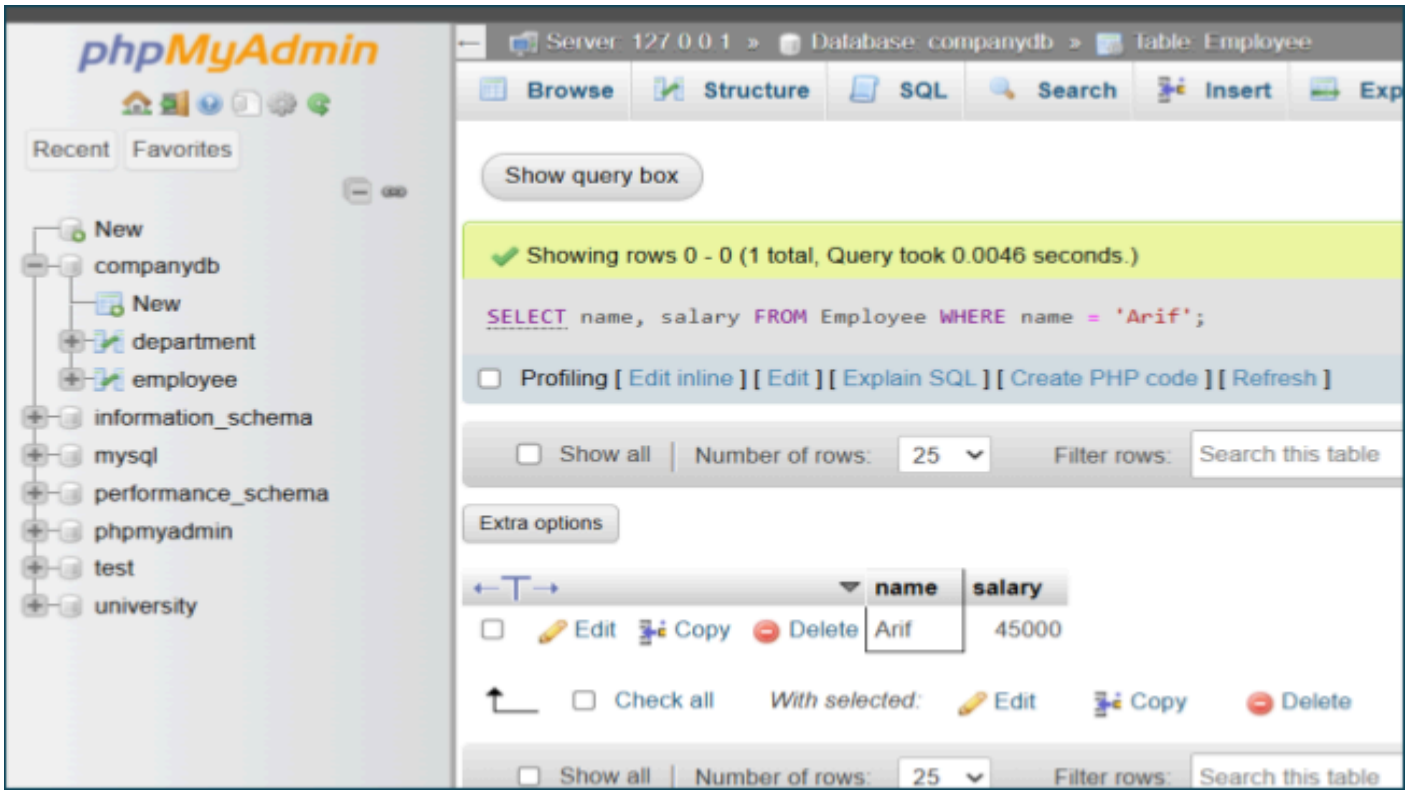


Problem 4:

Title: Find employees with name 'Arif'.

SQL Query: `SELECT name, salary FROM Employee WHERE name = 'Arif';`

Output:

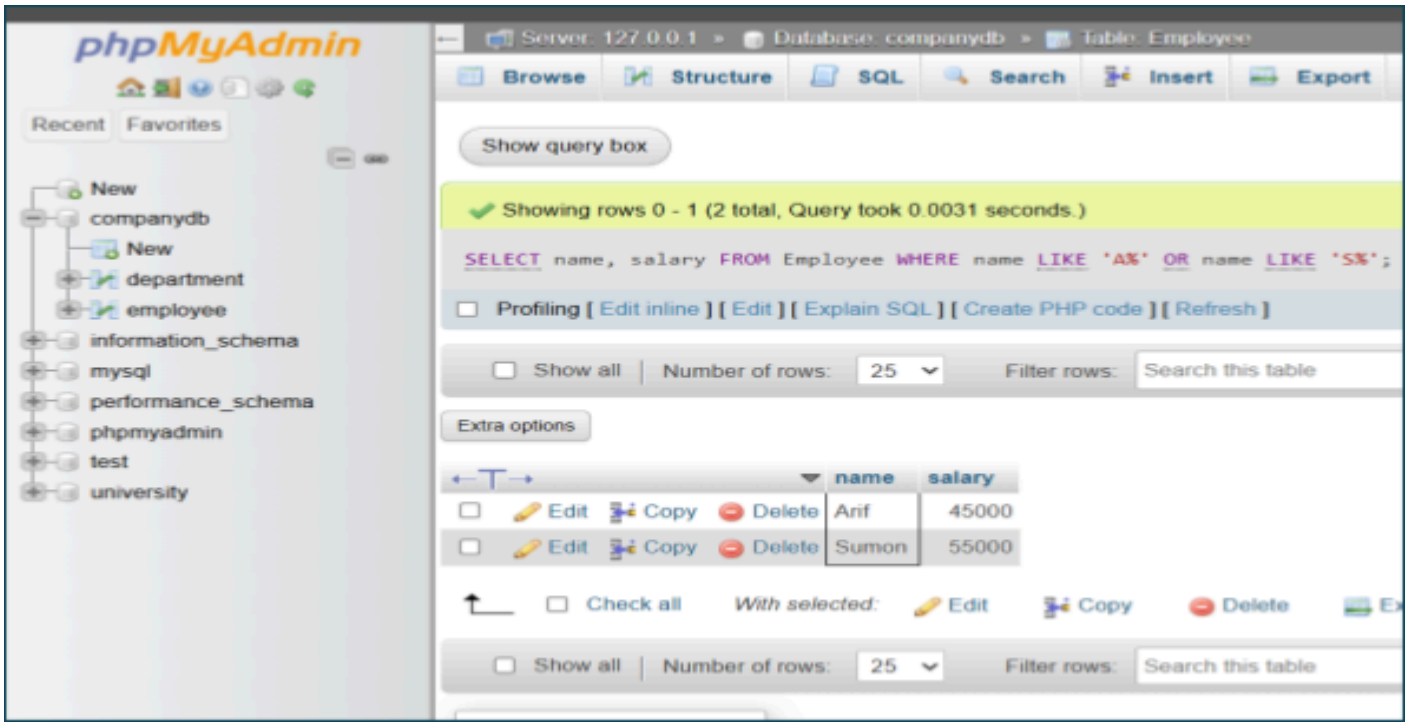


Problem 5:

Title: Find employees with names starting with 'A' or 'S'.

SQL Query: `SELECT name, salary FROM Employee WHERE name LIKE 'A%' OR name LIKE 'S%';`

Output:



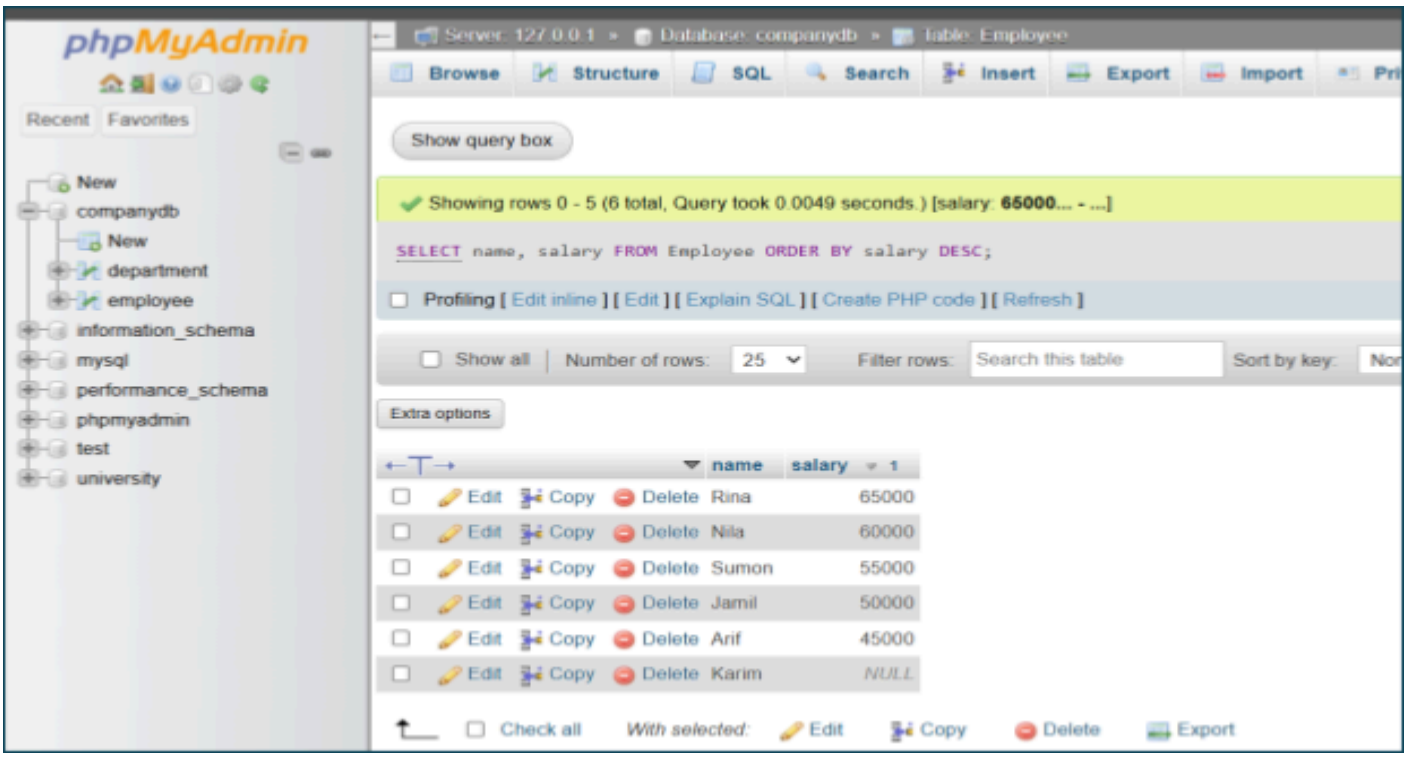
Problem 6:

Title: Sort employees by salary in descending order.

SQL Query:

```
SELECT name, salary FROM Employee ORDER BY salary DESC;
```

Output:

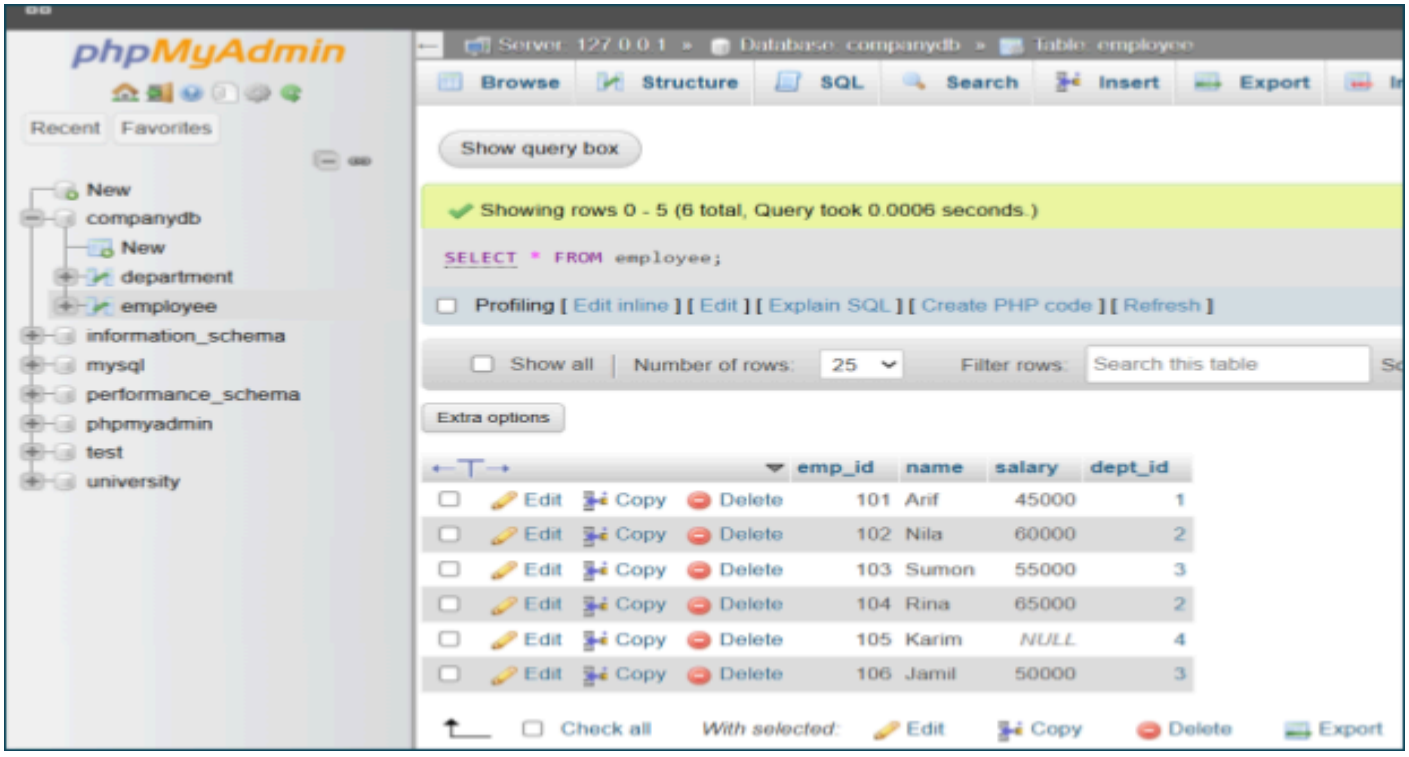


Problem 7:

Title: Display All Columns from Employee.

SQL Query: `SELECT * FROM employee;`

Output:

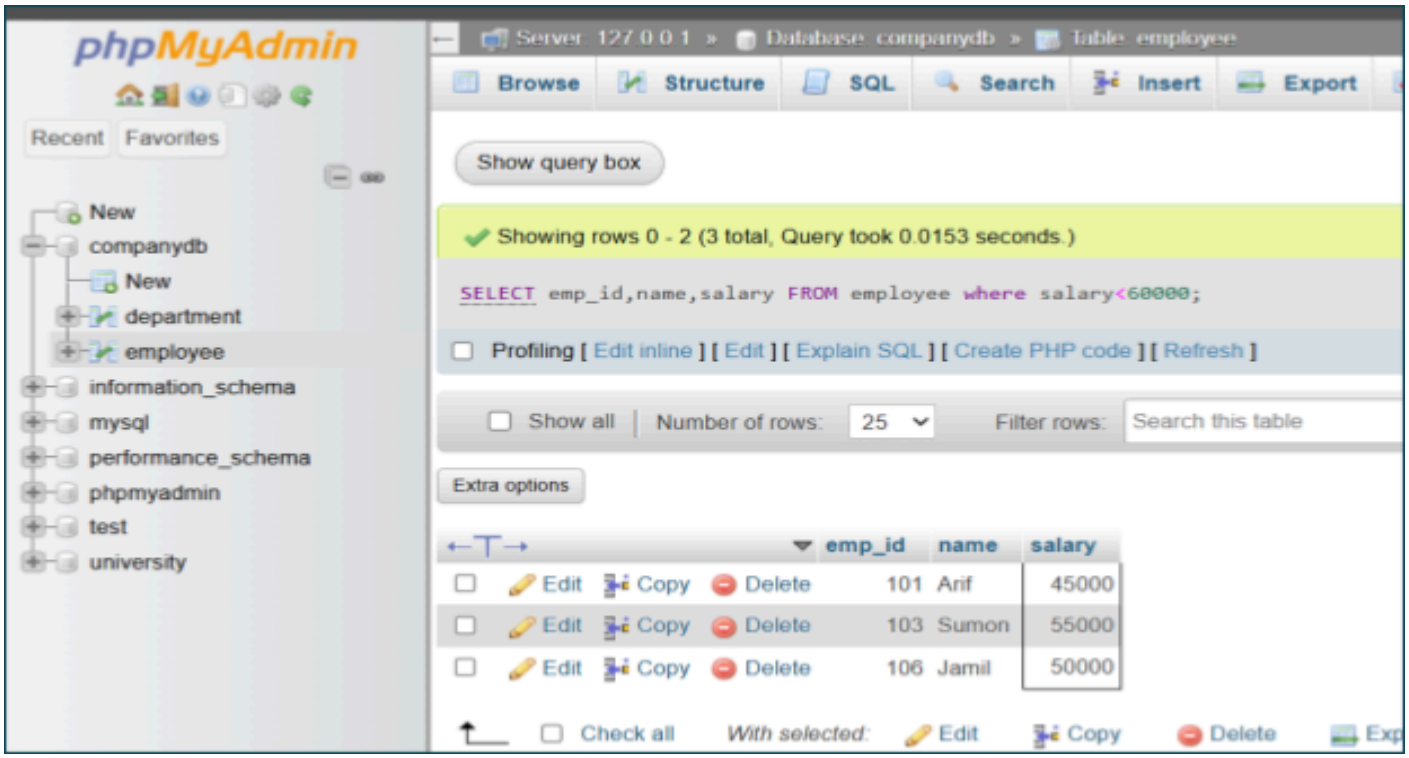


Problem 8:

Title: Find employees with a salary less than 60000.

SQL Query: `SELECT emp_id,name,salary FROM employee where salary<60000;`

Output:

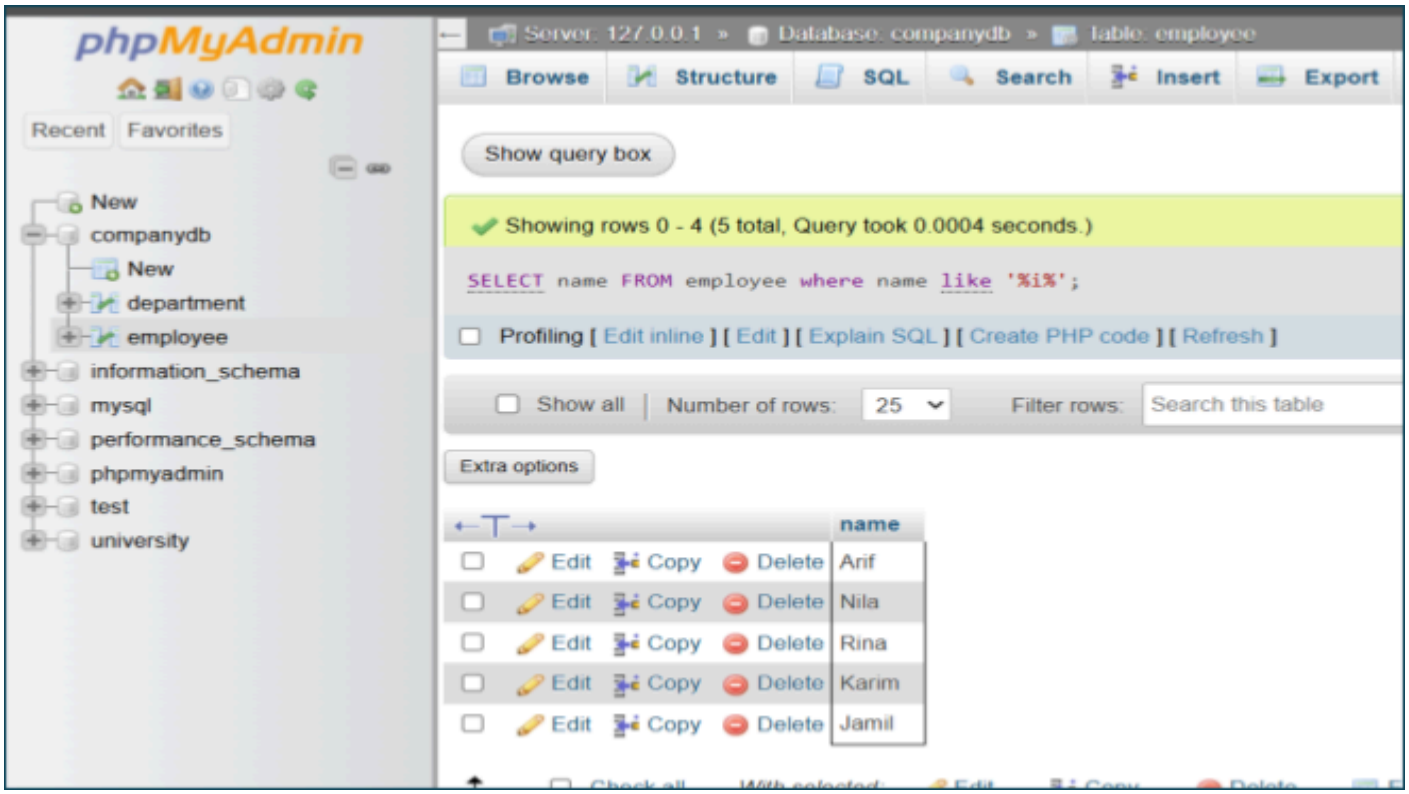


Problem 9:

Title: List employees with names containing 'i'.

SQL Query: `SELECT name FROM employee where name like '%i%';`

Output:

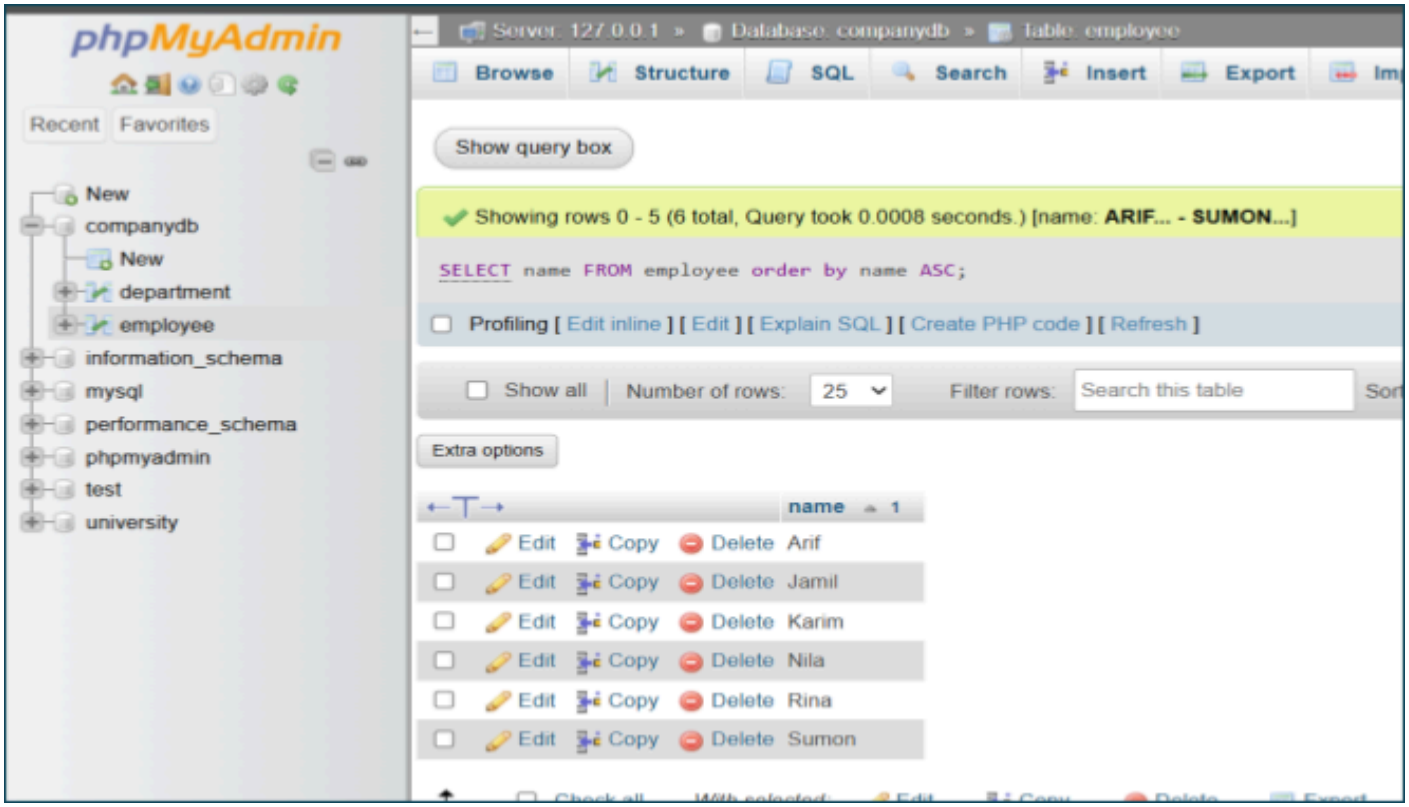


Problem 10:

Title: Sort employees by name in ascending order.

SQL Query: `SELECT name FROM employee order by name ASC;`

Output:

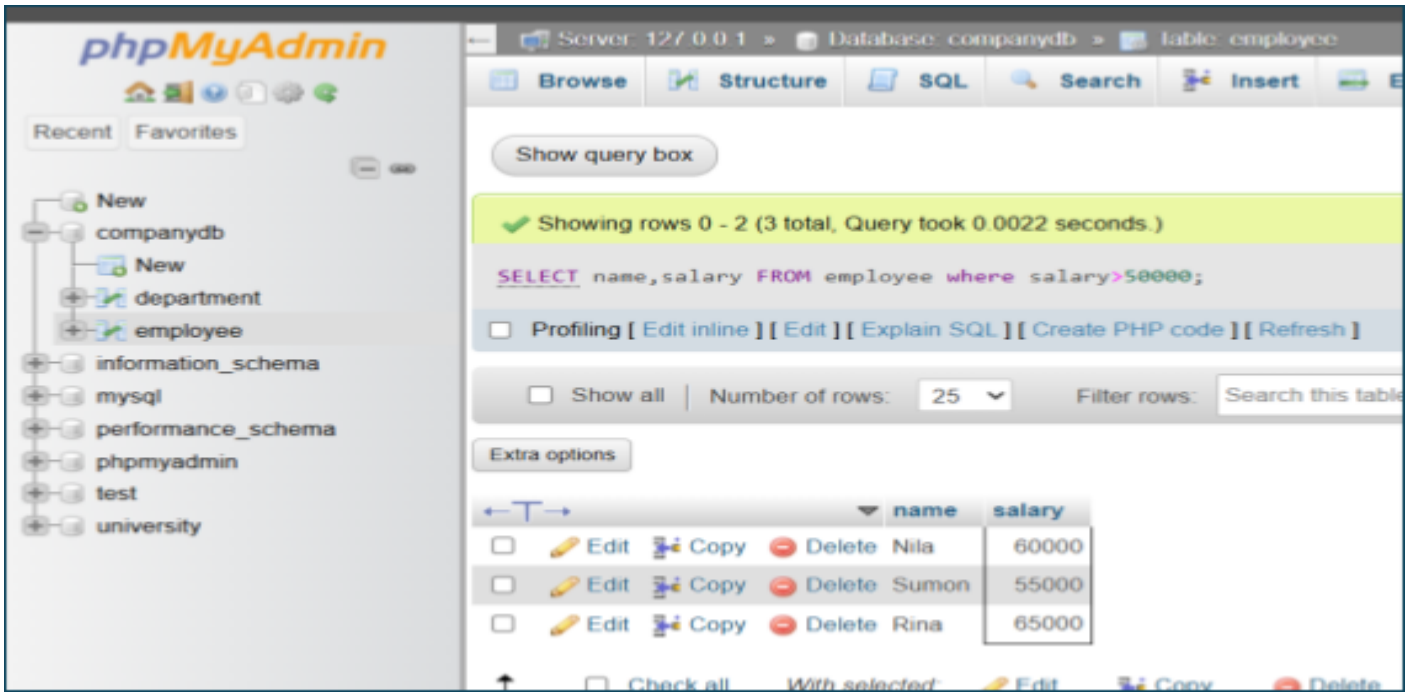


Problem 11:

Title: Find employees with a salary greater than 50000.

SQL Query: `SELECT name,salary FROM employee where salary>50000;`

Output:



Result: From this lab session, all SQL queries were successfully executed on the provided sample tables. The outputs show correct filtering, pattern matching, and ordering of data. The lab helped reinforce the use of conditional operators and basic SQL query structures.

Discussion: During this lab, I observed how SQL retrieves and filters data based on different conditions. Using operators such as <, >, and LIKE made it easier to extract specific records from the tables. Sorting data using ORDER BY helped understand how SQL organizes query results. A learning point was understanding how NULL behaves in conditional queries—for example, NULL does not satisfy comparison operations like < or >. Overall, the lab improved my understanding of data retrieval, filtering logic, and SQL query execution flow.

Conclusion: In this lab, I learned how to write fundamental SQL queries, filter data using conditional operators, apply pattern searches using LIKE, and sort data with ORDER BY. These skills are essential for performing data analysis and working with relational databases. The exercises strengthened my understanding of SQL fundamentals and query formulation.